Introduction to Data Science: Data pre-processing Tutorial 3

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1. Use for this question the data set `DataOutliers.csv`. Import the data set in R/Python and determine how many outliers this data set has. Explain your answer.

2. Use the data set `Inflation.csv` with monthly inflation (in %) in the Netherlands. Consider the following function f_λ with domain the positive real numbers:

$$f_\lambda(x) = \begin{cases} \frac{x^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0, \\ \log(x) & \text{if } \lambda = 0. \end{cases}$$

   Investigate the effect of λ when using the function f_λ to transform the inflation data.

3. Consider 2 observations, A and B in a 2-dimensional space and

$$A = (2, 1) \text{ and } B = (1, 2).$$

   (a) Determine the standardized versions of A and B and denote them by $(x_{1,1}, x_{1,2})$ and $(x_{2,1}, x_{2,2})$. The sample variance s^2 of a data set x_i for $i = 1, 2, \ldots, n$ can be determined as follows: $s^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$.

   (b) Determine the eigenvalues of the covariance matrix of the standardized data set $(x_{1,1}, x_{1,2})$ and $(x_{2,1}, x_{2,2})$.

   (c) Determine the principal components $\underline{\alpha}_1$ and $\underline{\alpha}_2$ of the data.

   (d) Show that the principal component are orthogonal.

   (e) The smallest eigenvalue is zero. Explain the intuition of this result.

Exam question

1. Consider a data set with p variables, denoted by $X_1, X_2, \ldots, X_p$. We have that each variable has mean 0 and the variance of X_i is σ^2. The first principal component Z_1 is given by

$$Z_1 = \sum_{i=1}^{n} \alpha_{1,i} X_i.$$

The vector $\underline{\alpha}_1$ is defined as follows:

$$\Sigma\underline{\alpha}_1 = \lambda_1\underline{\alpha}_1,$$

for some $\lambda_1 \in \mathbb{R}$.

Show that

$$\text{Corr}\left[Z_1, X_j\right] = \frac{\sqrt{\lambda_1}\alpha_{1,j}}{\sigma},$$

where λ_1 is the variance of Z_1.